

DOCKER ASSIGNMENT – ASSIGNMENT 5

Node.js Frontend + Flask Backend with Docker Compose

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Course: Tutedude – Full Stack Development

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GitHub Repository:

<https://github.com/Umair-bey/tutedude-assignments/tree/main/Assignment-5>

Docker Hub Images:

<https://hub.docker.com/r/umairkhan2026/flask-backend>

<https://hub.docker.com/r/umairkhan2026/node-frontend>

1. PROJECT OVERVIEW

This assignment demonstrates containerizing a two-service web application using Docker and Docker Compose.

The application consists of:

- A Node.js + Express frontend that serves an HTML form
- A Flask backend that receives form submissions and returns JSON responses
- A Docker Compose configuration that connects both services on a shared bridge network

When the user submits the form on the frontend (port 3000), the frontend forwards the data to the Flask backend (port 5000) using the Docker service name "backend" as the hostname. The backend processes the data and returns a JSON response, which is rendered on the frontend.

Key concepts demonstrated:

- Writing custom Dockerfiles for Node.js and Python applications
- Multi-container orchestration with docker-compose.yml
- Container-to-container communication over a user-defined bridge network
- Docker's internal DNS resolution using service names
- Publishing images to Docker Hub

2. FOLDER STRUCTURE

The project is organized with separate folders for the frontend and backend:

Assignment-5/

- ├── frontend/
 - │ ├── views/
 - │ └── form.ejs
 - │ ├── app.js
 - │ ├── package.json
 - │ └── Dockerfile
- ├── backend/
 - │ ├── app.py
 - │ ├── requirements.txt
 - │ └── Dockerfile
- ├── docker-compose.yml
- └── .gitignore

The screenshot shows a VS Code editor window with the following components:

- Explorer:** Displays the project structure of 'tutude-assignments'. It shows folders for Assignment-1 through Assignment-5. Under Assignment-5, there are folders for 'backend' and 'frontend'. The 'frontend' folder is expanded, showing 'form.ejs', 'app.js', 'Dockerfile', 'package.json', '.gitignore', and 'docker-compose.yml'. The 'README.md' file is selected.
- Editor:** Shows the content of 'app.js' in the 'frontend' folder. The code is as follows:

```
1 const express = require('express');
2 const bodyParser = require('body-parser');
3 const axios = require('axios');
4 const path = require('path');
5
6 const app = express();
7 app.set('view engine', 'ejs');
8 app.set('views', path.join(__dirname, 'views'));
9 app.use(bodyParser.urlencoded({ extended: true }));
10 app.use(bodyParser.json());
11
12 const BACKEND_URL = process.env.BACKEND_URL || 'http://backend:5000';
```
- Terminal:** Shows the output of a 'git log --oneline -5' command, which returns 'nothing to commit, working tree clean'. It also shows the output of a 'git status' command, which returns 'nothing to commit, working tree clean'. The terminal prompt is 'PS C:\Users\Umar khan\Desktop\tutude-assignments> git log --oneline -5'.

3. BACKEND (FLASK)

File: backend/app.py

The Flask application exposes two endpoints:

- GET / → Health check endpoint returning a JSON message
- POST /submit → Receives form data (name, email, message) and returns them as JSON

Flask-CORS is used to allow cross-origin requests from the frontend.

File: backend/requirements.txt

Flask==3.0.0

flask-cors==4.0.0

File: backend/Dockerfile

- Base image: python:3.11-slim
- Working directory: /app
- Copies requirements.txt and installs dependencies
- Exposes port 5000
- Runs: python app.py

4. FRONTEND (NODE.JS + EXPRESS)

File: frontend/app.js

The Express server:

- Uses EJS as the template engine
- Serves the HTML form at route "/"
- Handles POST /submit by forwarding the form data to the Flask backend
- Backend URL is taken from the environment variable BACKEND_URL (default: http://backend:5000)

File: frontend/views/form.ejs

An HTML form with three fields — Name, Email, Message — and a Submit button.

On successful submission, the JSON response from the backend is displayed in a success box.

File: frontend/package.json

Dependencies: express, ejs, axios, body-parser

File: frontend/Dockerfile

- Base image: node:18-alpine
- Working directory: /app
- Copies package.json and runs npm install
- Exposes port 3000
- Runs: npm start

5. DOCKER COMPOSE CONFIGURATION

File: docker-compose.yml

Two services are defined on a shared bridge network (app-network):

Service: frontend

- Builds from ./frontend
- Container name: frontend_service
- Port mapping: 3000:3000
- Environment: BACKEND_URL=<http://backend:5000>
- Depends on: backend

Service: backend

- Builds from ./backend
- Container name: backend_service
- Port mapping: 5000:5000

Networks:

- app-network (bridge driver)

Why this works:

Because both services share the same user-defined bridge network, Docker's internal DNS automatically resolves the service name "backend" to the backend container's IP. This lets the frontend reach the backend without hardcoding any IP address.

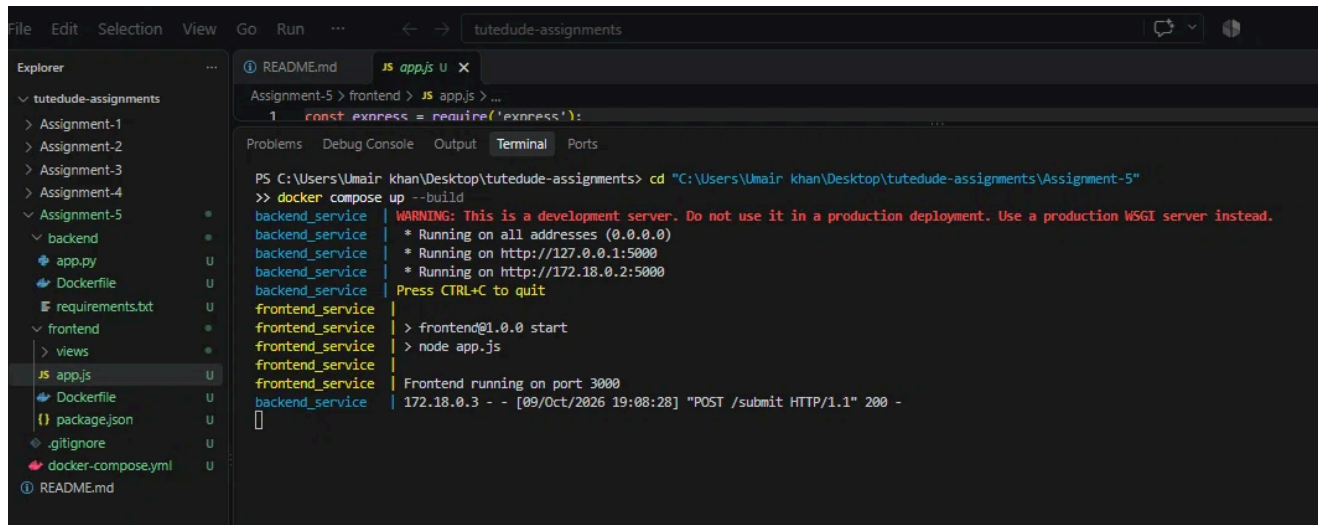
6. BUILD AND RUN COMMAND

From the Assignment-5 folder, run:

```
docker compose up --build
```

This command:

- Builds both Docker images from their Dockerfiles
- Creates the app-network bridge network
- Starts both containers



```
file Edit Selection View Go Run ... < -> tutedude-assignments

Explorer
└─ tutedude-assignments
  └─ Assignment-5
    └─ backend
      └─ app.py
      └─ Dockerfile
      └─ requirements.txt
      └─ frontend
        └─ views
        └─ JS app.js
        └─ Dockerfile
        └─ package.json
        └─ .gitignore
        └─ docker-compose.yml
        └─ README.md

Assignment-5 > frontend > JS app.js > ...
1 const express = require('express');

Problems Debug Console Output Terminal Ports

PS C:\Users\Umaisr\khan\Desktop\tutedude-assignments> cd "C:\Users\Umaisr\khan\Desktop\tutedude-assignments\Assignment-5"
>> docker compose up --build
backend_service | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
backend_service | * Running on all addresses (0.0.0.0)
backend_service | * Running on http://127.0.0.1:5000
backend_service | * Running on http://172.18.0.2:5000
backend_service | Press CTRL+C to quit
frontend_service |
frontend_service | > frontend@1.0.0 start
frontend_service | > node app.js
frontend_service | Frontend running on port 3000
backend_service | 172.18.0.3 - - [09/Oct/2026 19:08:28] "POST /submit HTTP/1.1" 200 -
[]
```

7. APPLICATION WORKING

Step 1: Open the browser at `http://localhost:3000` — the form loads.

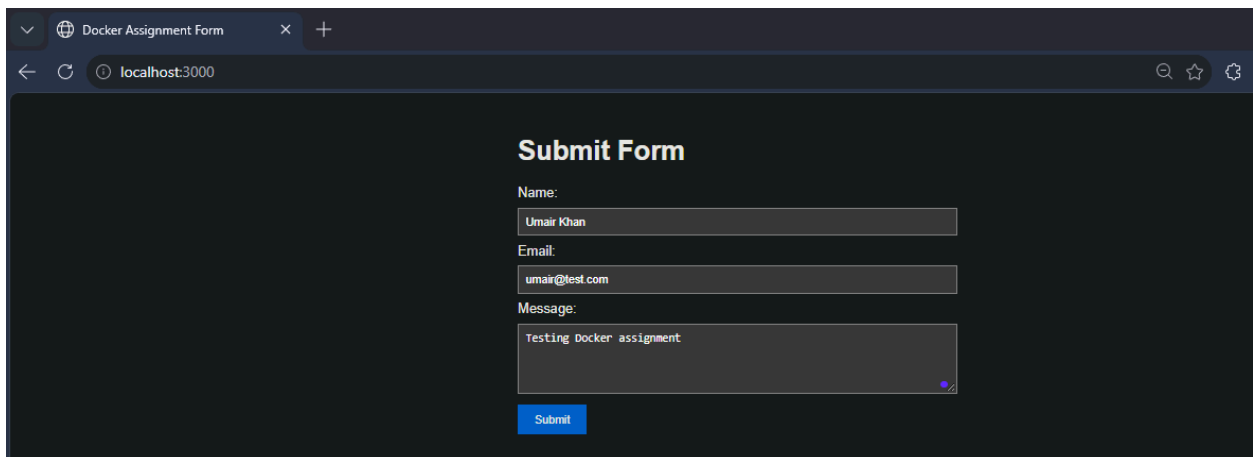
Step 2: Fill in Name, Email, and Message fields.

Step 3: Click Submit.

Step 4: The frontend forwards the POST request to the Flask backend.

Step 5: The backend returns a JSON response.

Step 6: The frontend renders the success box with the received data.

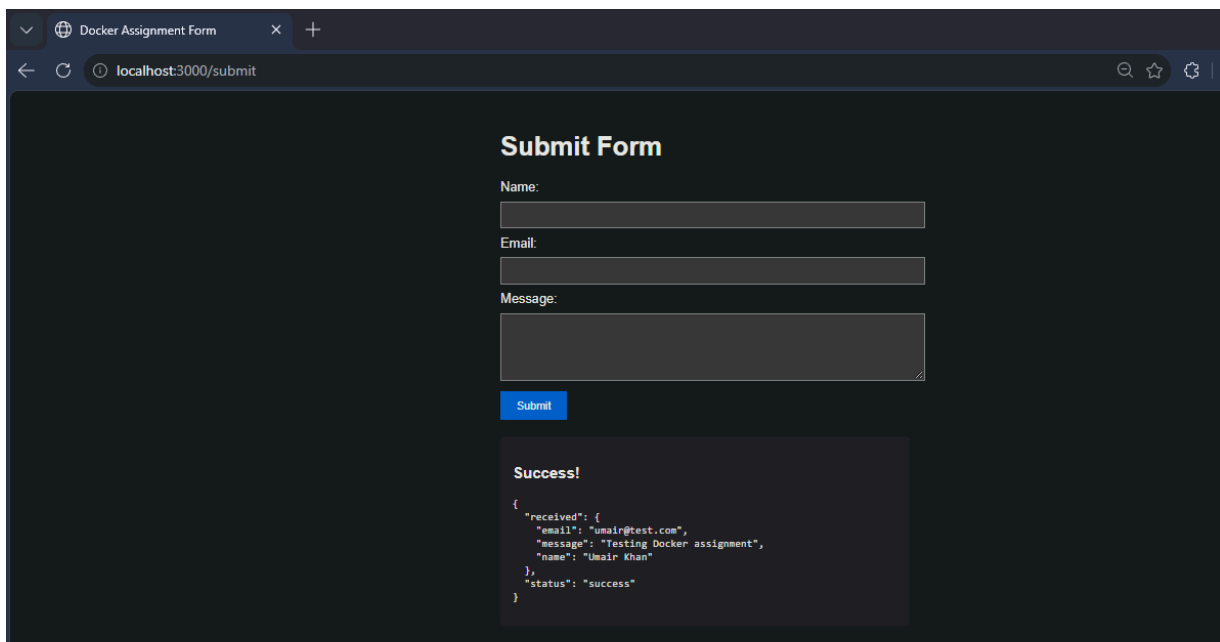


Submit Form

Name:

Email:

Message:



Submit Form

Name:

Email:

Message:

Success!

```
{
  "received": {
    "email": "umar@test.com",
    "message": "Testing Docker assignment",
    "name": "Umar Khan"
  },
  "status": "success"
}
```

8. RUNNING CONTAINERS AND IMAGES

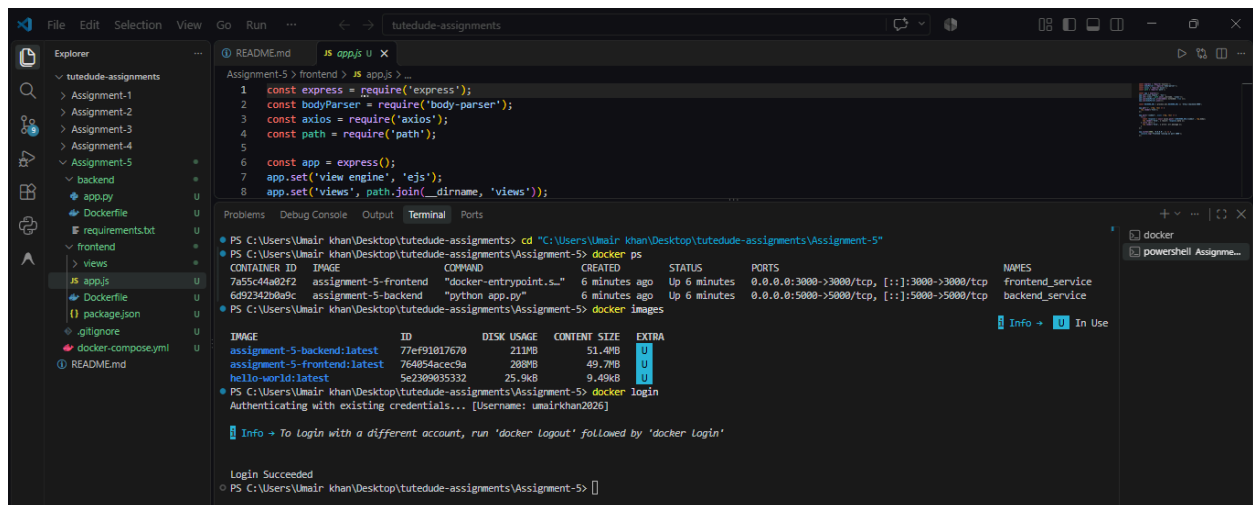
To verify that both containers are running:

```
docker ps
```

To list all locally built images:

```
docker images
```

Both containers are named `frontend_service` and `backend_service`, and are mapped to host ports 3000 and 5000 respectively.



The screenshot shows a Visual Studio Code editor with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The file explorer shows a project structure for 'tutodude-assignments' with files like 'app.js', 'Dockerfile', 'package.json', and 'README.md'. The code editor shows the content of 'app.js', which is a simple Express.js server. The terminal shows the output of the 'docker ps' command, listing two running containers: 'assignment-5-frontend' and 'assignment-5-backend'. Below the terminal output, there is a table showing the details of the Docker images.

```
PS C:\Users\Umaisir khan\Desktop\tutodude-assignments\Assignment-5> docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
7a55c44a02f2	assignment-5-frontend	"docker-entrypoint.s..."	6 minutes ago	Up 6 minutes	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp	frontend_service
6d92342b0bdc	assignment-5-backend	"python app.py"	6 minutes ago	Up 6 minutes	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	backend_service


```
PS C:\Users\Umaisir khan\Desktop\tutodude-assignments\Assignment-5> docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
assignment-5-backend:latest	77ef91017670	211MB	51.4MB	U
assignment-5-frontend:latest	764854acec9a	208MB	49.7MB	U
hello-world:latest	5e2309835332	25.9KB	9.49KB	U

PS C:\Users\Umaisir khan\Desktop\tutodude-assignments\Assignment-5> docker login
Authenticating with existing credentials... [Username: umaisirkhan2026]

Info -> To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
PS C:\Users\Umaisir khan\Desktop\tutodude-assignments\Assignment-5>

9. PUSHING IMAGES TO DOCKER HUB

The images were tagged and pushed to Docker Hub:

```
docker login
```

```
docker tag assignment-5-frontend umairkhan2026/node-frontend:latest
```

```
docker tag assignment-5-backend umairkhan2026/flask-backend:latest
```

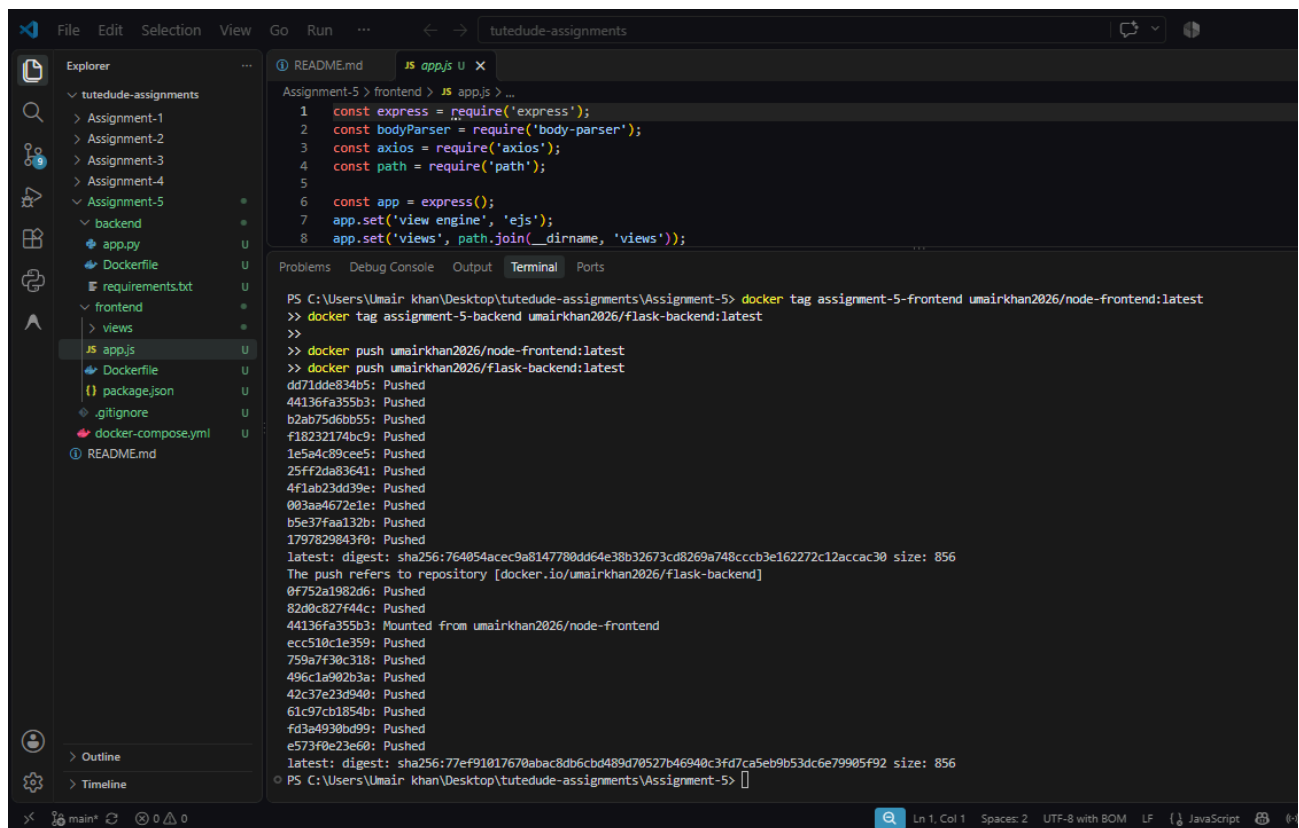
```
docker push umairkhan2026/node-frontend:latest
```

```
docker push umairkhan2026/flask-backend:latest
```

Both images are now publicly available:

• <https://hub.docker.com/r/umairkhan2026/node-frontend>

• <https://hub.docker.com/r/umairkhan2026/flask-backend>



The screenshot shows a Visual Studio Code editor with a project named 'tutedude-assignments'. The Explorer sidebar on the left shows the project structure, including folders for 'Assignment-1' through 'Assignment-5', and subfolders 'backend' and 'frontend'. The 'app.js' file is selected in the 'frontend' folder. The main editor area displays the code for 'app.js', which is a simple Express.js application. The bottom panel shows the 'Terminal' output, which contains the commands and output for pushing Docker images to Docker Hub. The output shows that the images were successfully pushed to the repository 'umairkhan2026' on Docker Hub.

```
Assignment-5 > frontend > JS app.js > ...
1 const express = require('express');
2 const bodyParser = require('body-parser');
3 const axios = require('axios');
4 const path = require('path');
5
6 const app = express();
7 app.set('view engine', 'ejs');
8 app.set('views', path.join(__dirname, 'views'));

Problems Debug Console Output Terminal Ports

PS C:\Users\Umar khan\Desktop\tutedude-assignments\Assignment-5> docker tag assignment-5-frontend umairkhan2026/node-frontend:latest
>> docker tag assignment-5-backend umairkhan2026/flask-backend:latest
>>
>> docker push umairkhan2026/node-frontend:latest
>> docker push umairkhan2026/flask-backend:latest
dd71dde834b5: Pushed
44136fa355b3: Pushed
b2ab75d6bb55: Pushed
f18232174bc9: Pushed
1e5a4c89cee5: Pushed
25ff2da83641: Pushed
4f1ab23dd39e: Pushed
003aa4672e1e: Pushed
b5e37faa132b: Pushed
1797829843f0: Pushed
latest: digest: sha256:764054acec9a8147780dd64e38b32673cd8269a748cccb3e162272c12accac30 size: 856
The push refers to repository [docker.io/umairkhan2026/flask-backend]
0f752a1982d6: Pushed
82d0c827f44c: Pushed
44136fa355b3: Mounted from umairkhan2026/node-frontend
ecc510c1e359: Pushed
759a7f30c318: Pushed
496c1a902b3a: Pushed
42c37e23d940: Pushed
61c97cb1854b: Pushed
fd3a4930bd99: Pushed
e573f0e23e60: Pushed
latest: digest: sha256:77ef91017670abac8db6cbd489d70527b46940c3fd7ca5eb9b53dc6e79905f92 size: 856
PS C:\Users\Umar khan\Desktop\tutedude-assignments\Assignment-5>
```

umairkhan2026/flask-backend

By [umairkhan2026](#) · Updated 1 minute ago

IMAGE

☆0 ↓0

Overview

Tags

Sort by

Newest

Filter tags

TAG

latest

Last pushed 1 minute by [umairkhan2026](#)

docker pull umairkhan2026/flask-backend:latest

Digest	OS/ARCH	Last pull	Compressed size
7c402252f66e	linux/amd64	less than 1 day	48.97 MB

umairkhan2026/node-frontend

By [umairkhan2026](#) · Updated 2 minutes ago

IMAGE

☆0 ↓0

Overview

Tags

Sort by

Newest

Filter tags

TAG

latest

Last pushed 2 minutes by [umairkhan2026](#)

docker pull umairkhan2026/node-frontend:latest

Digest	OS/ARCH	Last pull	Compressed size
825573a011c2	linux/amd64	less than 1 day	47.43 MB

10. GITHUB REPOSITORY

All source code, Dockerfiles, docker-compose.yml, and .gitignore were pushed to GitHub:

Repository: <https://github.com/Umair-bey/tutedude-assignments>

Assignment folder: <https://github.com/Umair-bey/tutedude-assignments/tree/main/Assignment-5>

The .gitignore file excludes node_modules/, __pycache__/, .vscode/, and other unnecessary files so that only the actual source code is committed.

The top screenshot shows the GitHub repository page for 'tutedude-assignments' by 'Umair-bey'. The repository is public and has 55 commits. The commit history shows a series of updates to the README.md file, with the most recent commit being 'Add Assignment-5: Docker with Node.js frontend and Flask backend' 2 minutes ago. The repository also has a README.md file and a .gitignore file.

The bottom screenshot shows a VS Code editor with the 'tutedude-assignments' project open. The Explorer pane on the left shows the project structure, including 'Assignment-1' through 'Assignment-5', 'backend', 'frontend', '.gitignore', 'docker-compose.yml', and 'README.md'. The main editor area shows the 'app.js' file in the 'frontend' directory, which contains the following code:

```
1 const express = require('express');
2 const bodyParser = require('body-parser');
3 const axios = require('axios');
4 const path = require('path');
5
6 const app = express();
7 app.set('view engine', 'ejs');
8 app.set('views', path.join(__dirname, 'views'));
```

The Output pane at the bottom shows the terminal output of the following commands:

```
PS C:\Users\Umair khan\Desktop\tutedude-assignments> git log --oneline -5
07b057e (HEAD -> main) Add Assignment-5: Docker with Node.js frontend and Flask backend
32488ad (origin/main, origin/HEAD) Update README.md
5032973 Merge remote README with local Assignment 4 additions
0409023 Task 4: Add final documentation (PDF + DOCX) and deployment screenshots
4807037 Update root README to include Assignment 4
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

nothing to commit, working tree clean
PS C:\Users\Umair khan\Desktop\tutedude-assignments> git push origin main
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 2 threads
Compressing objects: 100% (13/13), done.
Writing objects: 100% (15/15), 2.07 KiB | 163.00 KiB/s, done.
Total 15 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Umair-bey/tutedude-assignments.git
32488ad..07b057e main -> main
PS C:\Users\Umair khan\Desktop\tutedude-assignments>
```

11. CONCLUSION

Through this assignment, I successfully:

- Containerized a Node.js frontend and a Flask backend with separate Dockerfiles
- Connected both services using Docker Compose on a shared bridge network
- Enabled inter-container communication using Docker's internal DNS (service name as hostname)
- Deployed images to Docker Hub and pushed the source code to GitHub
- Applied best practices such as using .gitignore and environment variables

This assignment strengthened my understanding of containerization, multi-service architectures, and Docker networking.

